Leakage current estimation setup in Single MOS

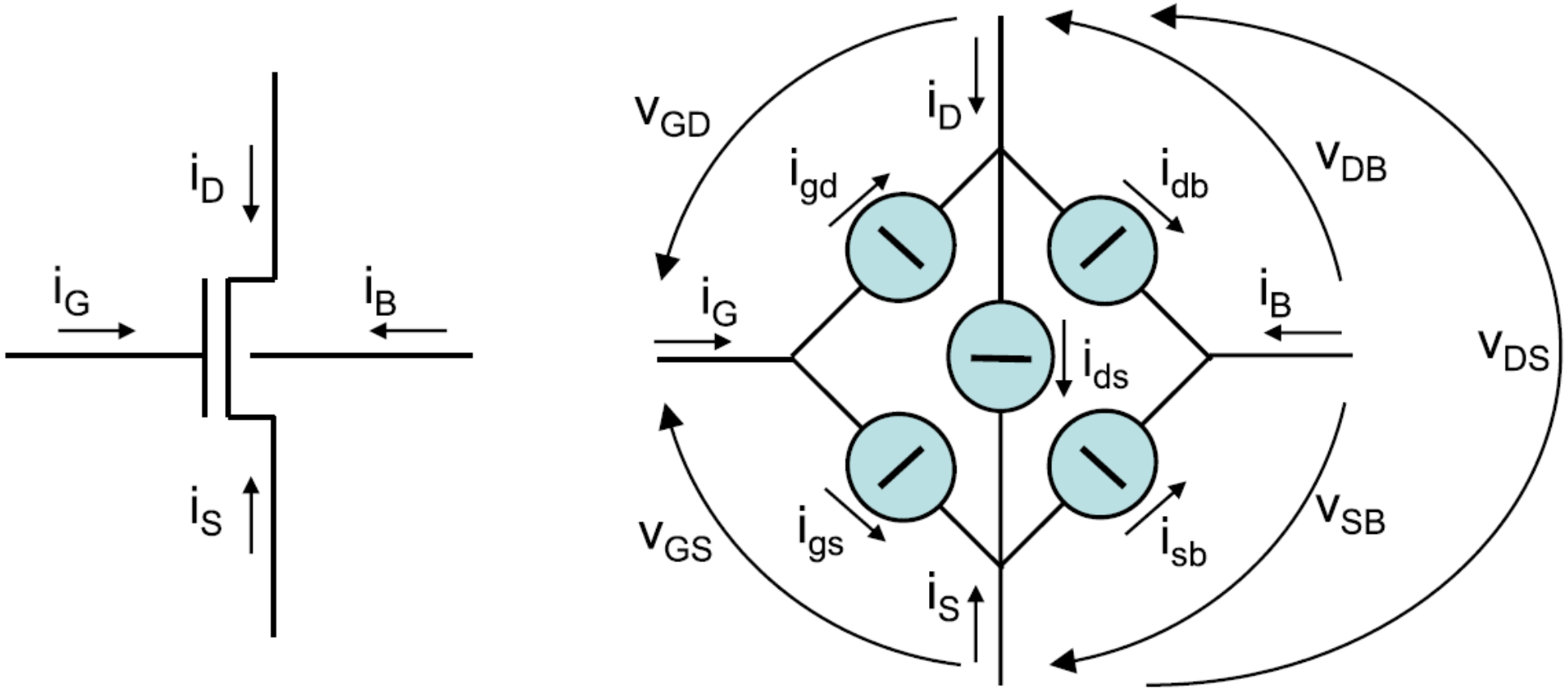


Fig. Abstract voltage-controlled non-linear bipole model of FET in static conditions, used in the proposed approach.

Leakage current estimation setup in Single MOS

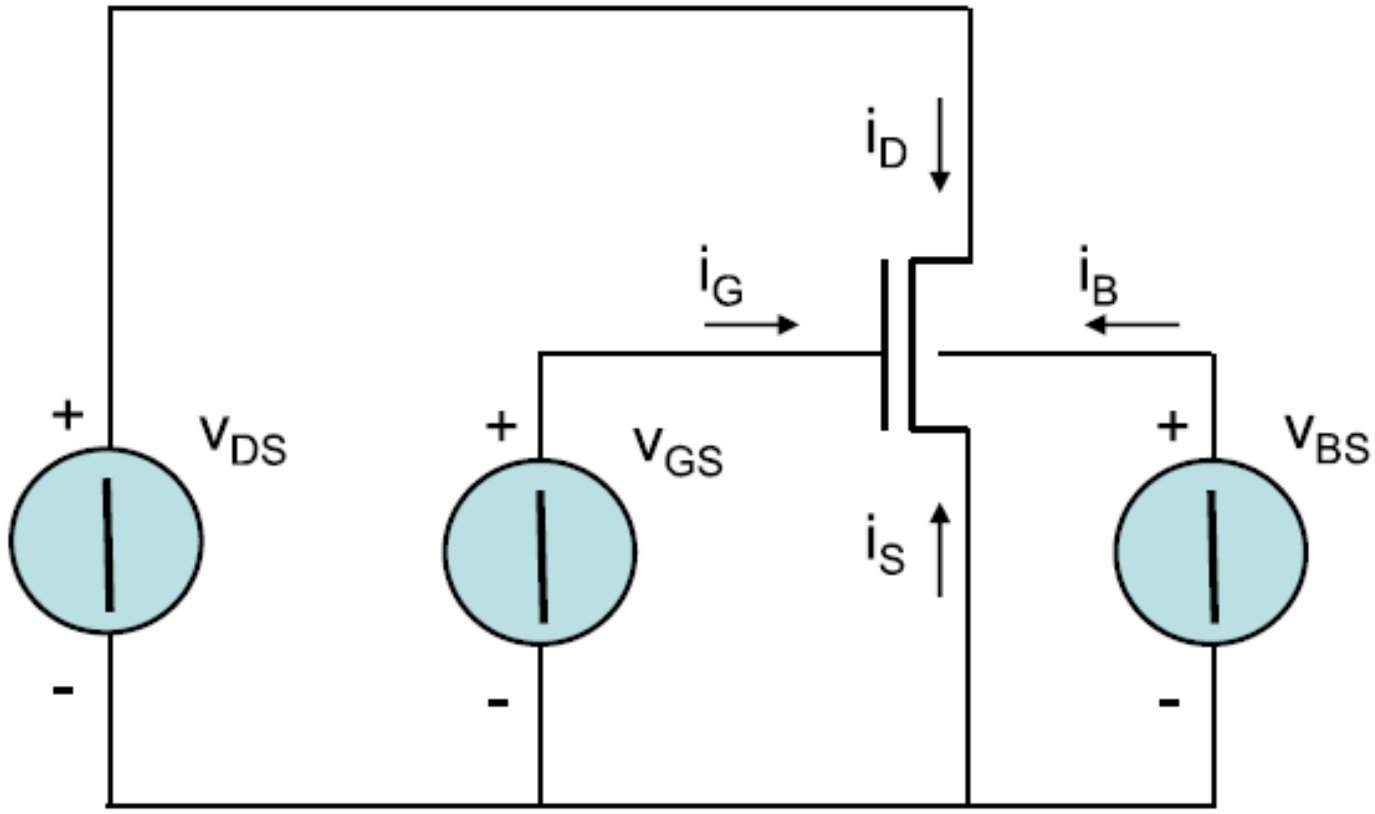
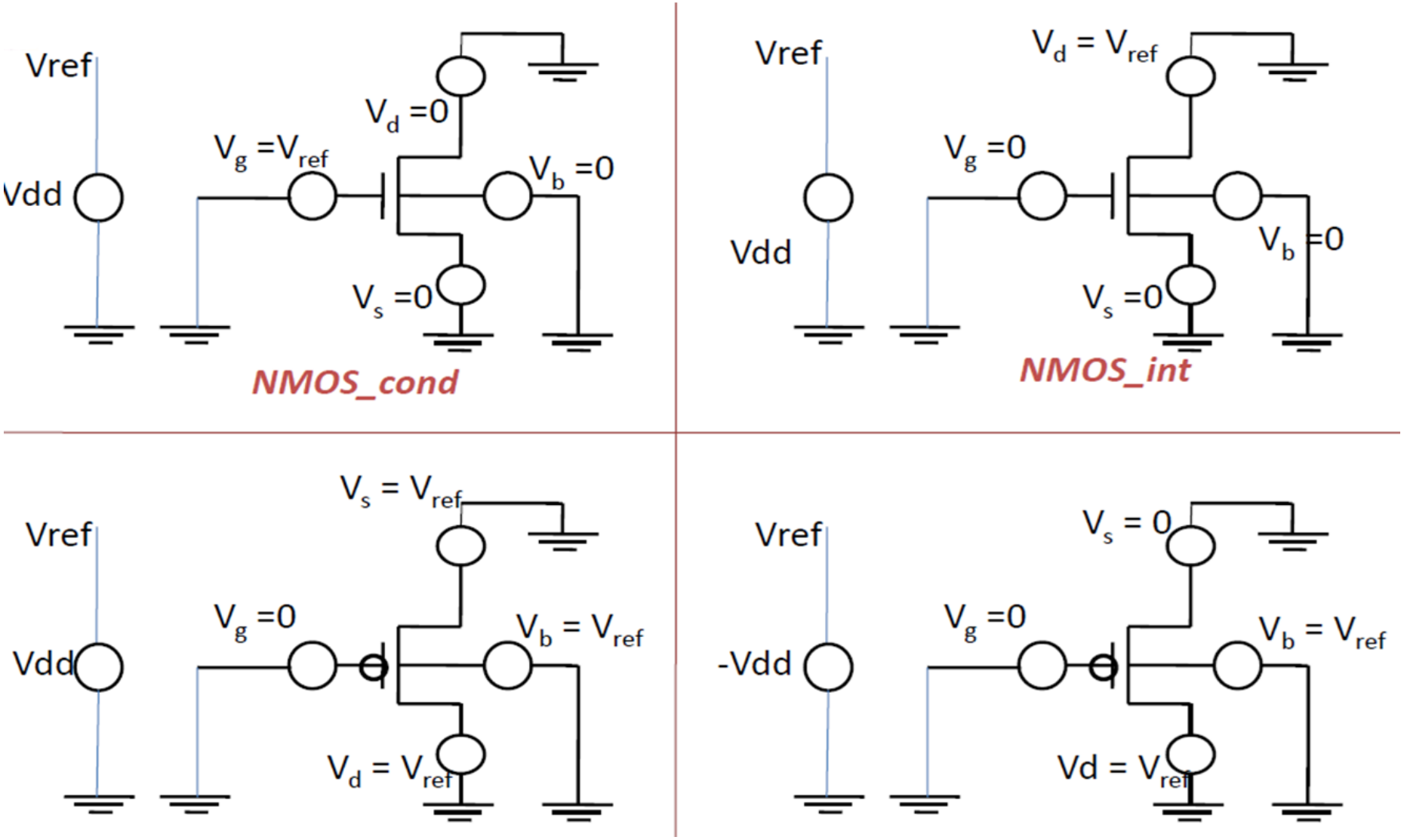


Fig. Simulation setup for the single device static current characterization (N-type devices. P-types are considered by complementary configuration).

Leakage current estimation setup in Single MOS



Sample_Database for Single_4MOS

-40°C ~ 125°C

Nom = 0.8V
0.81V - 0.94V

NFET_ON

PVT																		W, L		Terminal voltages				Leakage Currents				
temper	pvdd	nc0subn	nc0subp	nbodyp	tfinn	hfinp	toxpp	ni0subp	ni0subn	nbodyn	hfinn	eotp	nsdn	eotn	nsdp	toxpn	tfinp	lg	nfin	vd	vg	vs	vb	idrain	igate	isource	ibulk	leakage_current
7.00E+00	8.10E-01	2.77E+25	2.77E+25	1.03E+23	1.18E-08	2.61E-08	1.27E-09	1.13E+16	1.13E+16	1.03E+23	2.61E-08	7.93E-10	2.98E+26	7.93E-10	2.98E+26	1.27E-09	1.18E-08	1.96E-08	1.00E+00	0	8.10E-01	0	0	2.68E-11	-5.35E-11	2.68E-11	2.65E-17	-2.68E-11
2.20E+01	8.10E-01	2.92E+25	2.92E+25	9.81E+22	1.20E-08	2.51E-08	1.31E-09	1.09E+16	1.09E+16	9.81E+22	2.51E-08	8.20E-10	2.85E+26	8.20E-10	2.85E+26	1.31E-09	1.20E-08	2.03E-08	1.00E+00	0	8.10E-01	0	0	2.07E-11	-4.14E-11	2.07E-11	1.39E-16	-2.07E-11
3.20E+01	8.30E-01	2.87E+25	2.87E+25	1.01E+23	1.20E-08	2.61E-08	1.40E-09	9.89E+15	9.89E+15	1.01E+23	2.61E-08	8.22E-10	2.97E+26	8.22E-10	2.97E+26	1.40E-09	1.20E-08	2.06E-08	1.00E+00	0	8.30E-01	0	0	9.78E-12	-1.96E-11	9.78E-12	1.16E-15	-9.78E-12
8.60E+01	8.90E-01	2.84E+25	2.84E+25	9.49E+22	1.17E-08	2.56E-08	1.45E-09	1.13E+16	1.13E+16	9.49E+22	2.56E-08	8.19E-10	3.00E+26	8.19E-10	3.00E+26	1.45E-09	1.17E-08	2.04E-08	1.00E+00	0	8.90E-01	0	0	1.09E-11	-2.42E-11	1.09E-11	2.38E-12	-1.09E-11
9.60E+01	9.20E-01	2.97E+25	2.97E+25	1.00E+23	1.15E-08	2.49E-08	1.37E-09	1.16E+16	1.16E+16	1.00E+23	2.49E-08	7.87E-10	2.85E+26	7.87E-10	2.85E+26	1.37E-09	1.15E-08	2.07E-08	1.00E+00	0	9.20E-01	0	0	3.20E-11	-8.63E-11	3.20E-11	2.22E-11	-3.20E-11
9.90E+01	7.70E-01	2.82E+25	2.82E+25	9.34E+22	1.21E-08	2.56E-08	1.34E-09	1.09E+16	1.09E+16	9.34E+22	2.56E-08	8.42E-10	2.99E+26	8.42E-10	2.99E+26	1.34E-09	1.21E-08	1.92E-08	1.00E+00	0	7.70E-01	0	0	2.11E-11	-4.21E-11	2.11E-11	5.02E-15	-2.11E-11
-2.40E+01	8.40E-01	2.72E+25	2.72E+25	1.04E+23	1.22E-08	2.70E-08	1.34E-09	1.12E+16	1.12E+16	1.04E+23	2.70E-08	8.37E-10	3.06E+26	8.37E-10	3.06E+26	1.34E-09	1.22E-08	2.06E-08	1.00E+00	0	8.40E-01	0	0	1.22E-11	-2.44E-11	1.22E-11	7.95E-18	-1.22E-11
1.08E+02	8.00E-01	2.98E+25	2.98E+25	1.04E+23	1.13E-08	2.65E-08	1.32E-09	1.13E+16	1.13E+16	1.04E+23	2.65E-08	7.96E-10	2.88E+26	7.96E-10	2.88E+26	1.32E-09	1.13E-08	1.97E-08	1.00E+00	0	8.00E-01	0	0	3.62E-11	-7.25E-11	3.62E-11	4.80E-14	-3.62E-11
-1.00E+00	8.80E-01	2.89E+25	2.89E+25	9.74E+22	1.22E-08	2.64E-08	1.36E-09	1.06E+16	1.06E+16	9.74E+22	2.64E-08	8.22E-10	2.98E+26	8.22E-10	2.98E+26	1.36E-09	1.22E-08	2.04E-08	1.00E+00	0	8.80E-01	0	0	1.41E-11	-2.82E-11	1.41E-11	1.59E-15	-1.41E-11
6.00E+01	8.00E-01	2.89E+25	2.89E+25	9.98E+22	1.21E-08	2.54E-08	1.33E-09	1.11E+16	1.11E+16	9.98E+22	2.54E-08	8.54E-10	2.83E+26	8.54E-10	2.83E+26	1.33E-09	1.21E-08	2.05E-08	1.00E+00	0	8.00E-01	0	0	2.17E-11	-4.33E-11	2.17E-11	2.10E-15	-2.17E-11
8.50E+01	7.90E-01	2.77E+25	2.77E+25	1.01E+23	1.19E-08	2.47E-08	1.39E-09	1.12E+16	1.12E+16	1.01E+23	2.47E-08	7.61E-10	3.12E+26	7.61E-10	3.12E+26	1.39E-09	1.19E-08	1.95E-08	1.00E+00	0	7.90E-01	0	0	1.08E-11	-2.15E-11	1.08E-11	3.55E-15	-1.08E-11
7.30E+01	8.40E-01	2.77E+25	2.77E+25	9.95E+22	1.23E-08	2.54E-08	1.29E-09	1.10E+16	1.10E+16	9.95E+22	2.54E-08	7.82E-10	3.04E+26	7.82E-10	3.04E+26	1.29E-09	1.23E-08	1.88E-08	1.00E+00	0	8.40E-01	0	0	4.30E-11	-8.61E-11	4.30E-11	3.18E-14	-4.30E-11
3.10E+01	9.20E-01	3.02E+25	3.02E+25	9.68E+22	1.24E-08	2.63E-08	1.31E-09	1.15E+16	1.15E+16	9.68E+22	2.63E-08	7.81E-10	3.18E+26	7.81E-10	3.18E+26	1.31E-09	1.24E-08	2.02E-08	1.00E+00	0	9.20E-01	0	0	3.95E-11	-7.92E-11	3.95E-11	2.78E-13	-3.95E-11
7.40E+01	8.80E-01	2.89E+25	2.89E+25	1.07E+23	1.19E-08	2.55E-08	1.36E-09	1.17E+16	1.17E+16	1.07E+23	2.55E-08	7.79E-10	3.01E+26	7.79E-10	3.01E+26	1.36E-09	1.19E-08	2.02E-08	1.00E+00	0	8.80E-01	0	0	2.53E-11	-5.11E-11	2.53E-11	4.61E-13	-2.53E-11
1.14E+02	7.80E-01	2.81E+25	2.81E+25	1.00E+23	1.15E-08	2.52E-08	1.37E-09	9.91E+15	9.91E+15	1.00E+23	2.52E-08	8.03E-10	2.98E+26	8.03E-10	2.98E+26	1.37E-09	1.15E-08	1.94E-08	1.00E+00	0	7.80E-01	0	0	1.65E-11	-3.30E-11	1.65E-11	1.81E-14	-1.65E-11
9.40E+01	8.50E-01	2.72E+25	2.72E+25	9.59E+22	1.22E-08	2.67E-08	1.50E-09	1.06E+16	1.06E+16	9.59E+22	2.67E-08	7.85E-10	3.10E+26	7.85E-10	3.10E+26	1.50E-09	1.22E-08	1.88E-08	1.00E+00	0	8.50E-01	0	0	4.88E-12	-9.98E-12	4.88E-12	2.20E-13	-4.88E-12
-1.40E+01	9.00E-01	3.05E+25	3.05E+25	9.60E+22	1.24E-08	2.60E-08	1.31E-09	1.12E+16	1.12E+16	9.60E+22	2.60E-08	7.97E-10	2.96E+26	7.97E-10	2.96E+26	1.31E-09	1.24E-08	1.94E-08	1.00E+00	0	9.00E-01	0	0	2.34E-11	-4.68E-11	2.34E-11	1.39E-15	-2.34E-11
-4.00E+00	9.20E-01	2.78E+25	2.78E+25	1.04E+23	1.20E-08	2.70E-08	1.31E-09	9.99E+15	9.99E+15	1.04E+23	2.70E-08	8.09E-10	3.13E+26	8.09E-10	3.13E+26	1.31E-09	1.20E-08	2.05E-08	1.00E+00	0	9.20E-01	0	0	2.82E-11	-5.64E-11	2.82E-11	1.60E-14	-2.82E-11
4.60E+01	8.40E-01	2.95E+25	2.95E+25	1.00E+23	1.22E-08	2.61E-08	1.28E-09	1.09E+16	1.09E+16	1.00E+23	2.61E-08	7.58E-10	2.90E+26	7.58E-10	2.90E+26	1.28E-09	1.22E-08	2.15E-08	1.00E+00	0	8.40E-01	0	0	4.30E-11	-8.61E-11	4.30E-11	6.25E-15	-4.30E-11
7.50E+01	8.10E-01	2.82E+25	2.82E+25	9.67E+22	1.16E-08	2.56E-08	1.32E-09	1.02E+16	1.02E+16	9.67E+22	2.56E-08	7.83E-10	3.00E+26	7.83E-10	3.00E+26	1.32E-09	1.16E-08	1.98E-08	1.00E+00	0	8.10E-01	0	0	2.60E-11	-5.20E-11	2.60E-11	7.81E-15	-2.60E-11
6.50E+01	8.00E-01	2.80E+25	2.80E+25	1.02E+23	1.22E-08	2.50E-08	1.36E-09	1.08E+16	1.08E+16	1.02E+23	2.50E-08	8.52E-10	2.96E+26	8.52E-10	2.96E+26	1.36E-09	1.22E-08	1.98E-08	1.00E+00	0	8.00E-01	0	0	1.61E-11	-3.21E-11	1.61E-11	2.51E-15	-1.61E-11
-3.50E+01	9.30E-01	2.77E+25	2.77E+25	1.01E+23	1.19E-08	2.62E-08	1.41E-09	1.09E+16	1.09E+16	1.01E+23	2.62E-08	8.25E-10	2.98E+26	8.25E-10	2.98E+26	1.41E-09	1.19E-08	2.04E-08	1.00E+00	0	9.30E-01	0	0	7.64E-12	-1.53E-11	7.64E-12	1.74E-15	-7.64E-12
8.00E+00	8.70E-01	2.87E+25	2.87E+25	1.01E+23	1.16E-08	2.73E-08	1.31E-09	1.07E+16	1.07E+16	1.01E+23	2.73E-08	8.27E-10	2.96E+26	8.27E-10	2.96E+26	1.31E-09	1.16E-08	2.05E-08	1.00E+00	0	8.70E-01	0	0	2.64E-11	-5.27E-11	2.64E-11	2.00E-15	-2.64E-11
1.06E+02	7.70E-01	2.85E+25	2.85E+25	1.05E+23	1.22E-08	2.58E-08	1.40E-09	1.09E+16	1.09E+16	1.05E+23	2.58E-08	8.38E-10	2.84E+26	8.38E-10	2.84E+26	1.40E-09	1.22E-08	2.07E-08	1.00E+00	0	7.70E-01	0	0	1.29E-11	-2.58E-11	1.29E-11	8.08E-15	-1.29E-11
1.02E+02	7.90E-01	2.68E+25	2.68E+25	9.43E+22	1.23E-08	2.59E-08	1.38E-09	1.11E+16	1.11E+16	9.43E+22	2.59E-08	8.18E-10	3.07E+26	8.18E-10	3.07E+26	1.38E-09	1.23E-08	2.05E-08	1.00E+00	0	7.90E-01	0	0	1.56E-11	-3.12E-11	1.56E-11	1.56E-14	-1.56E-11
2.00E+01	8.60E-01	2.97E+25	2.97E+25	9.66E+22	1.24E-08	2.67E-08	1.31E-09	1.07E+16	1.07E+16	9.66E+22	2.67E-08	7.69E-10	2.93E+26	7.69E-10	2.93E+26	1.31E-09	1.24E-08	1.93E-08	1.00E+00	0	8.60E-01	0	0	2.61E-11	-5.21E-11	2.61E-11	2.07E-15	-2.61E-11
1.20E+01	8.80E-01	2.85E+25	2.85E+25	9.88E+22	1.19E-08	2.62E-08	1.44E-09	1.13E+16	1.13E+16	9.88E+22	2.62E-08	7.79E-10	2.97E+26	7.79E-10	2.97E+26	1.44E-09	1.19E-08	1.96E-08	1.00E+00	0	8.80E-01	0	0	5.77E-12	-1.16E-11	5.77E-12	4.24E-15	-5.78E-12
9.30E+01	7.60E-01	2.80E+25	2.80E+25	9.47E+22	1.18E-08	2.72E-08	1.26E-09	1.14E+16	1.14E+16	9.47E+22	2.72E-08	7.85E-10	2.99E+26	7.85E-10	2.99E+26	1.26E-09	1.18E-08	1.92E-08	1.00E+00	0	7.60E-01	0	0	4.94E-11	-9.88E-11	4.94E-11	1.86E-15	-4.94E-11
9.70E+01	7.80E-01	2.82E+25	2.82E+25	9.78E+22	1.20E-08	2.59E-08	1.25E-09	1.11E+16	1.11E+16	9.78E+22	2.59E-08	8.06E-10	2.76E+26	8.06E-10	2.76E+26	1.25E-09	1.20E-08	2.02E-08	1.00E+00	0	7.80E-01	0	0	6.34E-11	-1.27E-10	6.34E-11	8.31E-15	-6.34E-11
9.00E+01	8.40E-01	2.90E+25	2.90E+25	1.09E+23	1.22E-08	2.58E-08	1.32E-09	1.07E+16	1.07E+16	1.09E+23	2.58E-08	7.94E-10	3.16E+26	7.94E-10	3.16E+26	1.32E-09	1.22E-08	2.05E-08	1.00E+00	0	8.40E-01	0	0	3.89E-11	-7.79E-11	3.89E-11	1.36E-13	-3.89E-11

- Develop M/L model

$R2 > 0.99$

Err% < 1% compared with simulation

Stacks: 2 and 3 intermediate node voltages

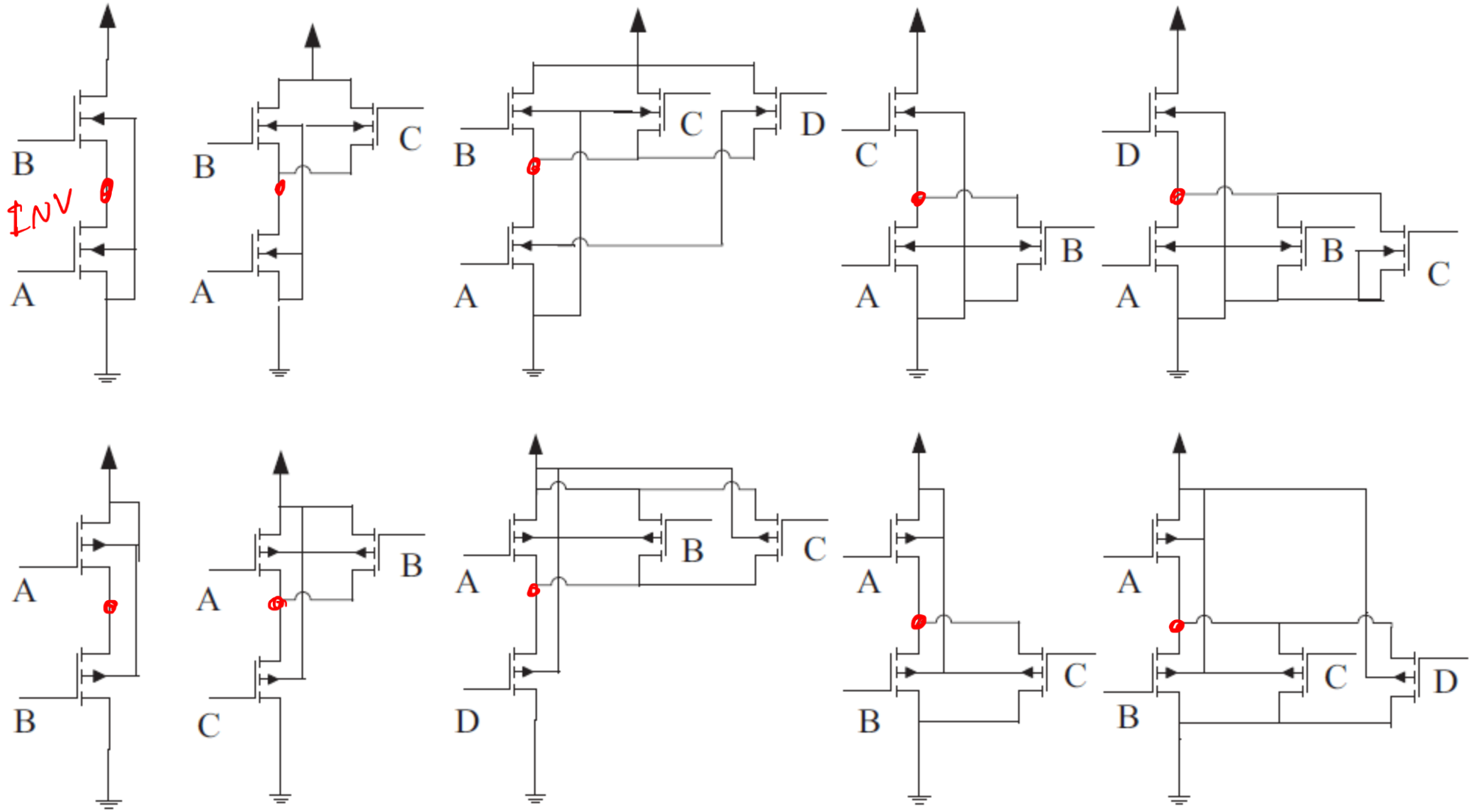


Fig. Internal node voltage circuits for two N-type and two P-type stacked devices (only one internal node).

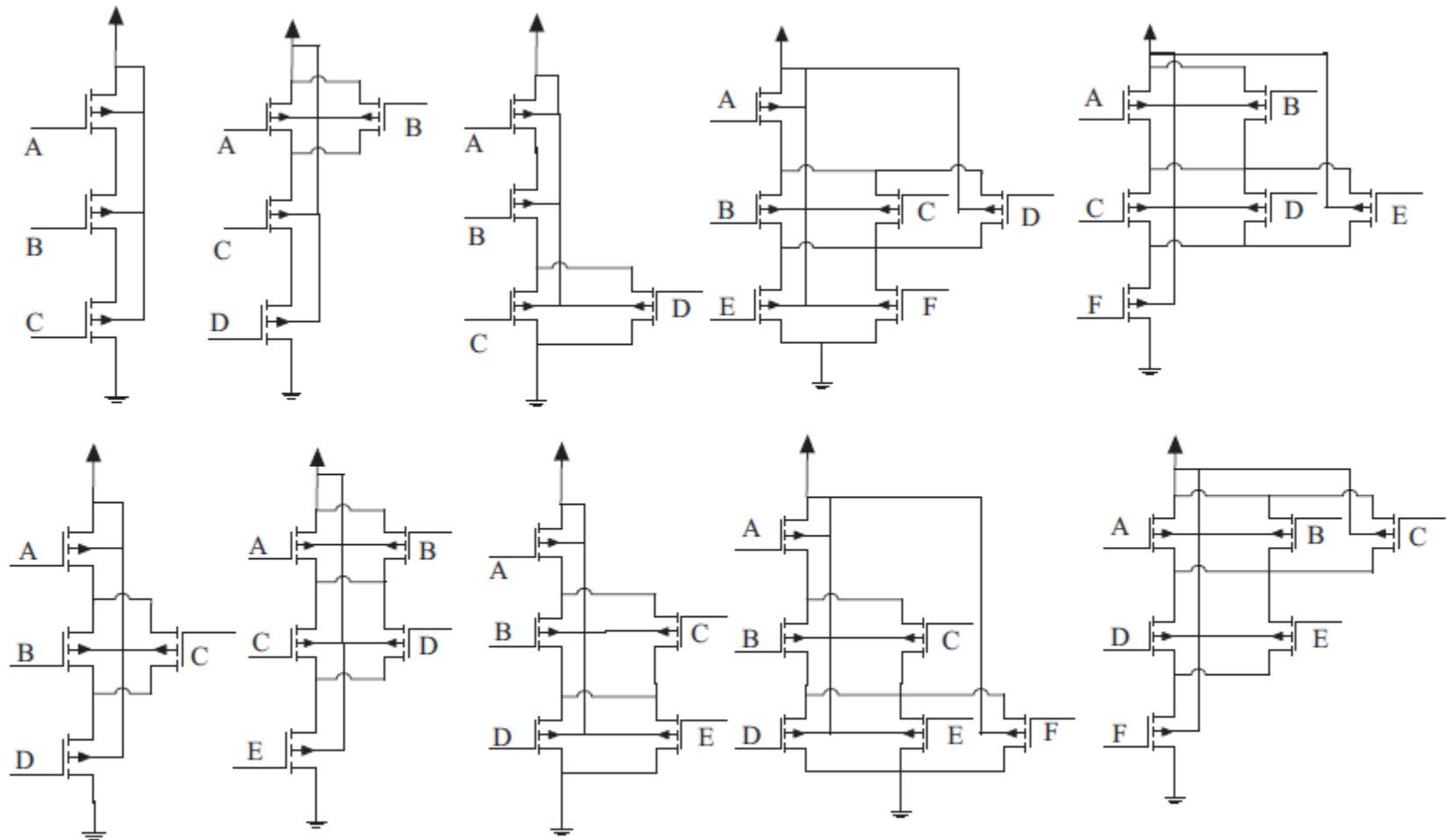
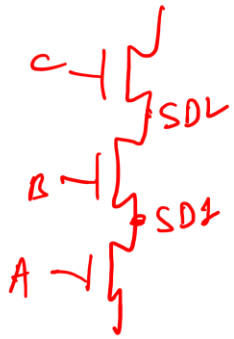


Fig. Internal node voltage circuits for three N-type and two P-type stacked devices (two internal nodes). NMOS stacks are considered accordingly.

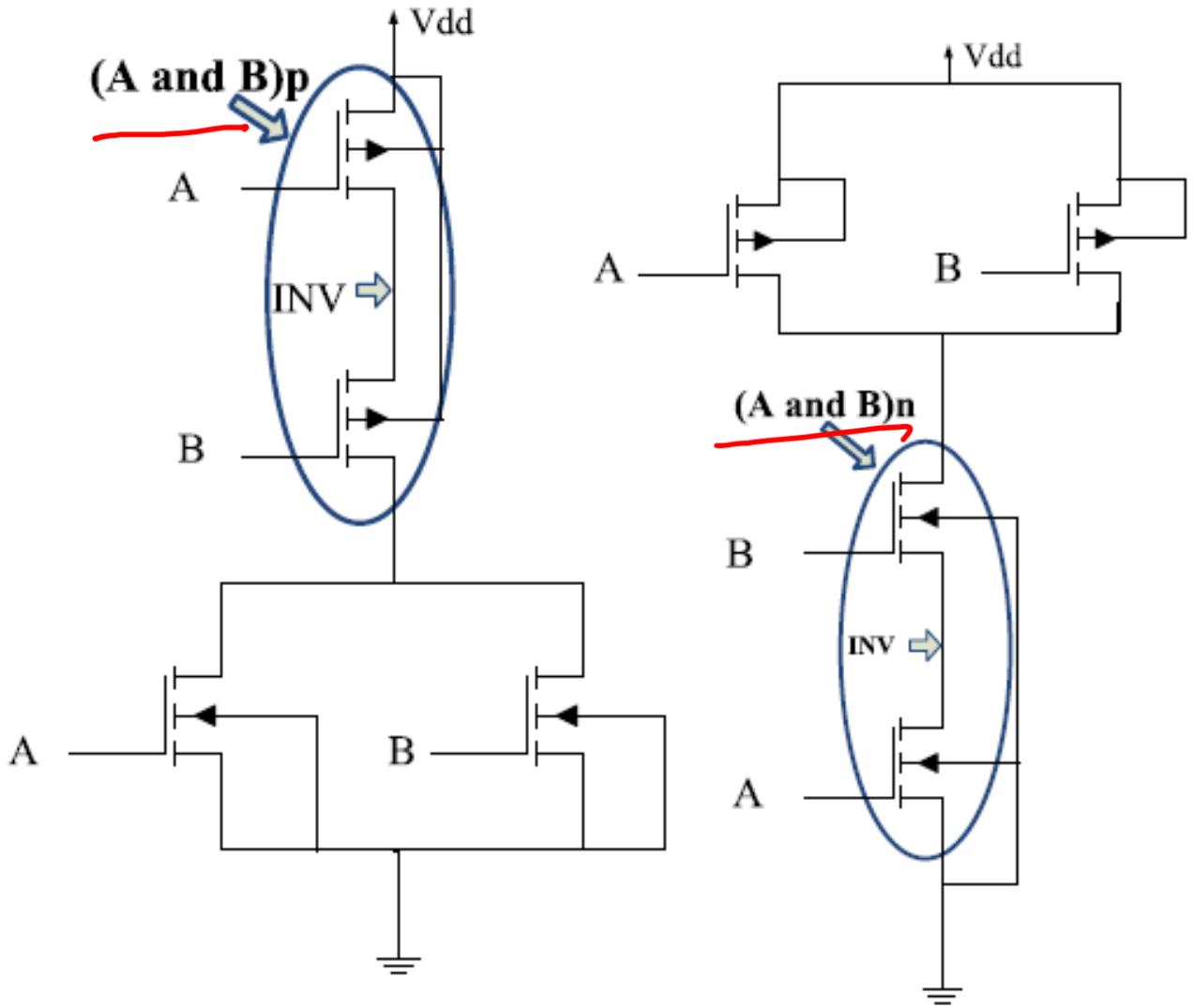
INV_Package

```
Constant matrix_20nm_V09_T30: matrix_23NP:= (
-- AandBandCn
--V(SD1)     V(SD2)             Supply at 30C
(8.042e-02, 1.492e-02,  9.000e-01), --0
(8.851e-01, 6.488e-02,  9.000e-01), --1
(6.678e-02, 6.678e-02,  9.000e-01), --2
(8.377e-01, 8.229e-01,  9.000e-01), --3
(6.585e-02, 3.883e-06,  9.000e-01), --4
(8.368e-01, 3.087e-05,  9.000e-01), --5
(7.139e-05, 3.578e-05,  9.000e-01), --6
(3.470e-01, 1.699e-01,  9.000e-01), --7

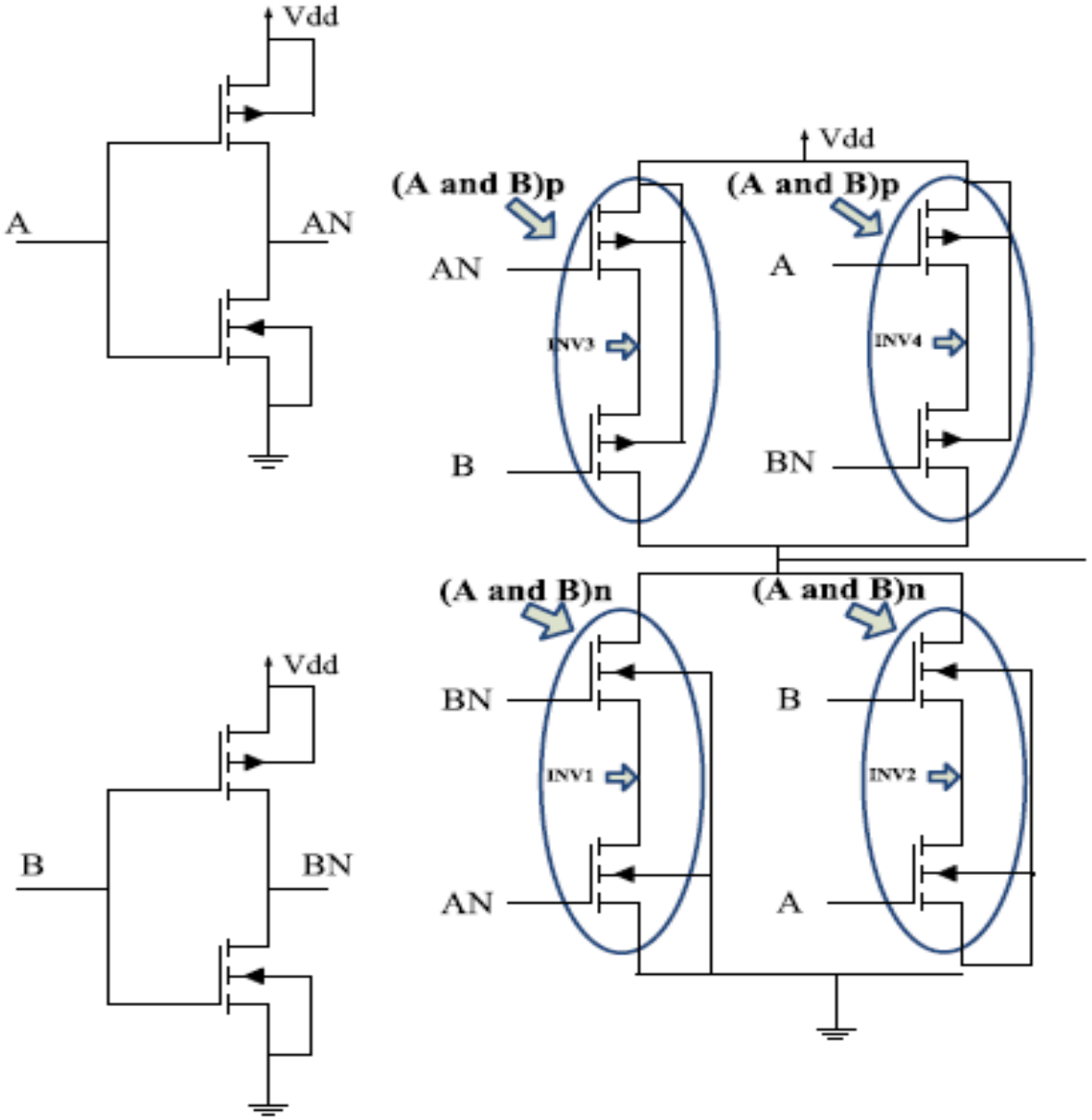
-- AandB_and_CorDn
--V(SD1)     V(SD2)             Supply at 30
(1.578e-02, 9.961e-02,  9.000e-01), --8
(6.488e-02, 8.851e-01,  9.000e-01), --9
(6.488e-02, 8.851e-01,  9.000e-01), --10
(6.533e-02, 8.920e-01,  9.000e-01), --11
(8.505e-02, 8.505e-02,  9.000e-01), --12
(8.229e-01, 8.377e-01,  9.000e-01), --13
(8.229e-01, 8.377e-01,  9.000e-01), --14
(8.305e-01, 8.558e-01,  9.000e-01), --15
(4.129e-06, 8.426e-02,  9.000e-01), --16
(3.087e-05, 8.368e-01,  9.000e-01), --17
(3.087e-05, 8.368e-01,  9.000e-01), --18
(3.213e-05, 8.544e-01,  9.000e-01), --19
(7.113e-05, 1.421e-04,  9.000e-01), --20
(1.699e-01, 3.470e-01,  9.000e-01), --21
(1.699e-01, 3.470e-01,  9.000e-01), --22
(2.090e-01, 4.464e-01,  9.000e-01), --23
```



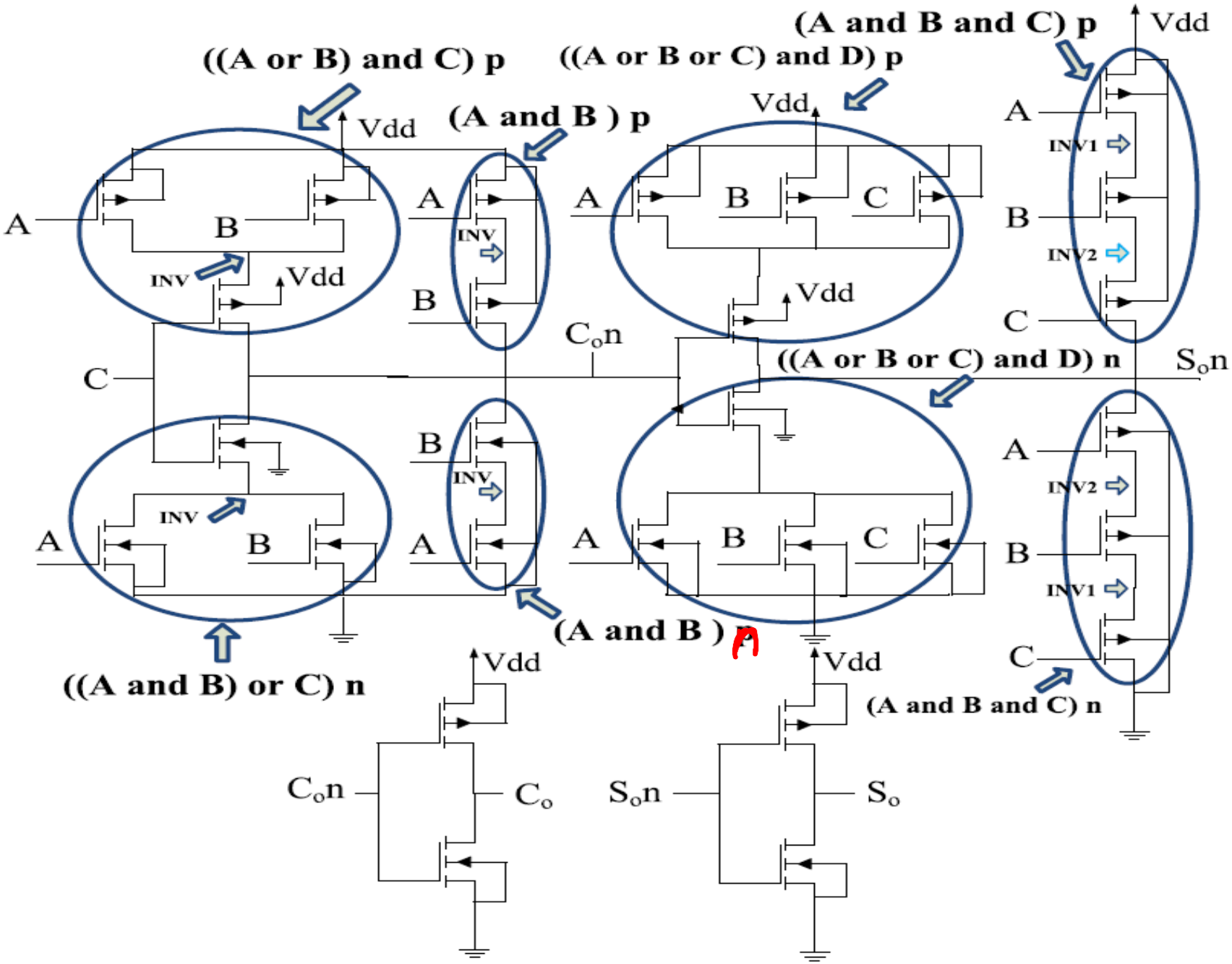
Example



Example


Vdd
A
AN
Vdd
(A and B)p
(A and B)p
AN
A
INV3
INV4
B
BN
(A and B)n
(A and B)n
Vdd
BN
B
INV1
INV2
B
BN
AN
A

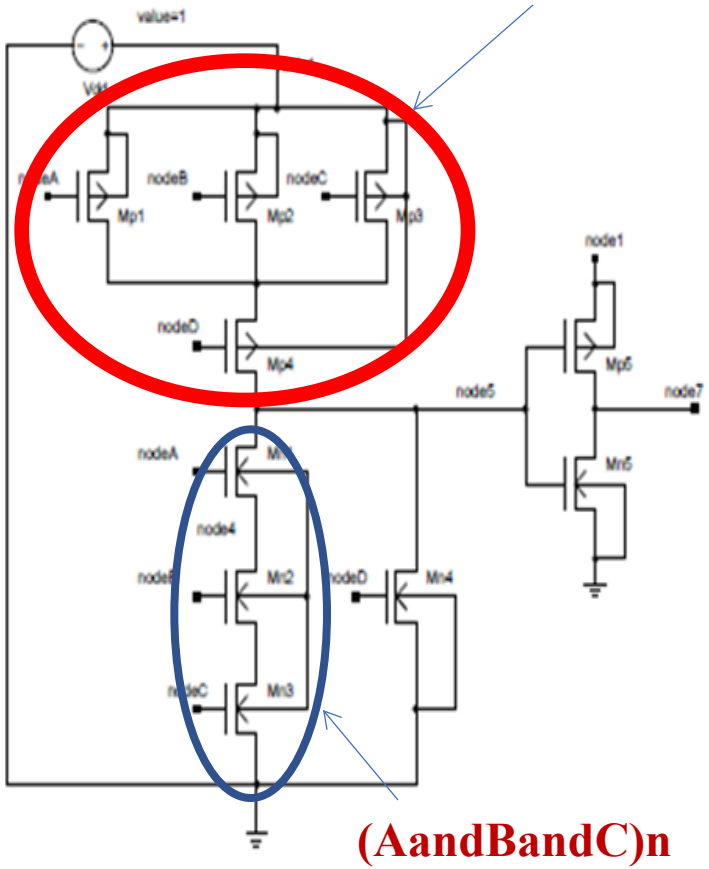
Example


((A or B) and C) p
(A and B) p
((A or B or C) and D) p
(A and B and C) p
Vdd
A
B
C
INV
INV1
INV2
Con
((A or B or C) and D) n
Son
((A and B) or C) n
(A and B) n
(A and B and C) n
Co
So

Example- Cell AO31

Recognizing pull-up and pull-down stacks

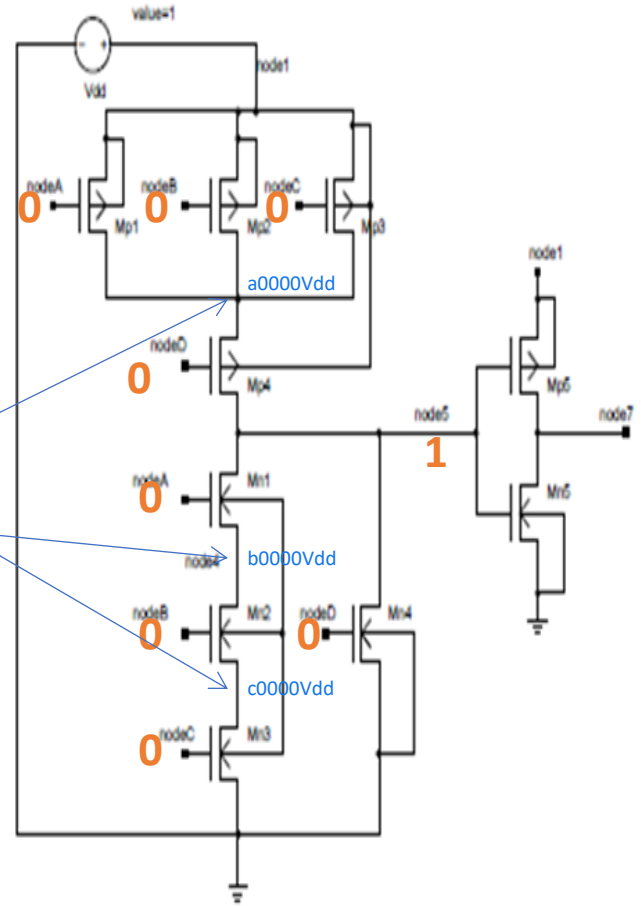
((AorBorC)andD)p



(AandBandC)n

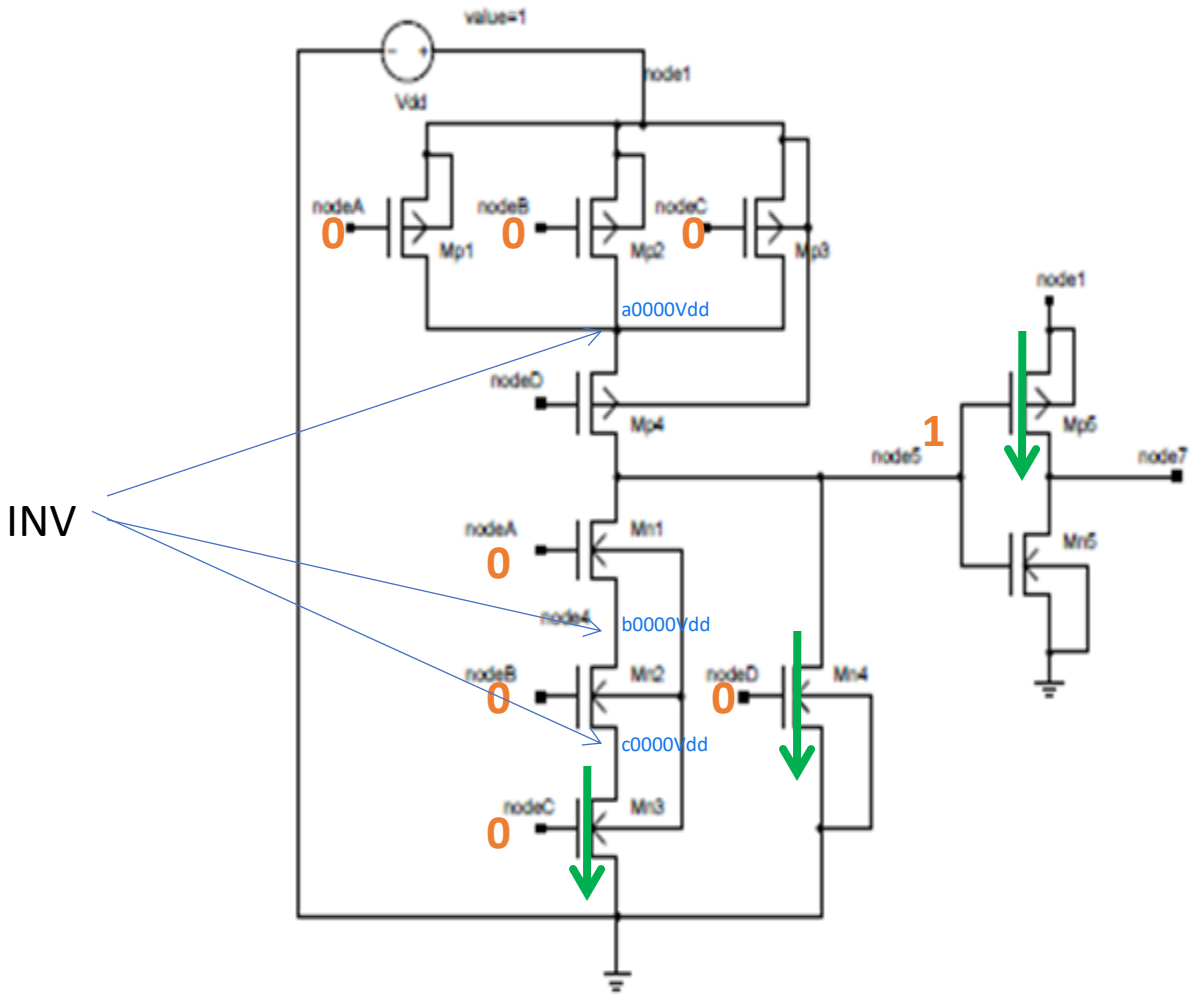
Finding Intermediate nodes

IN
V



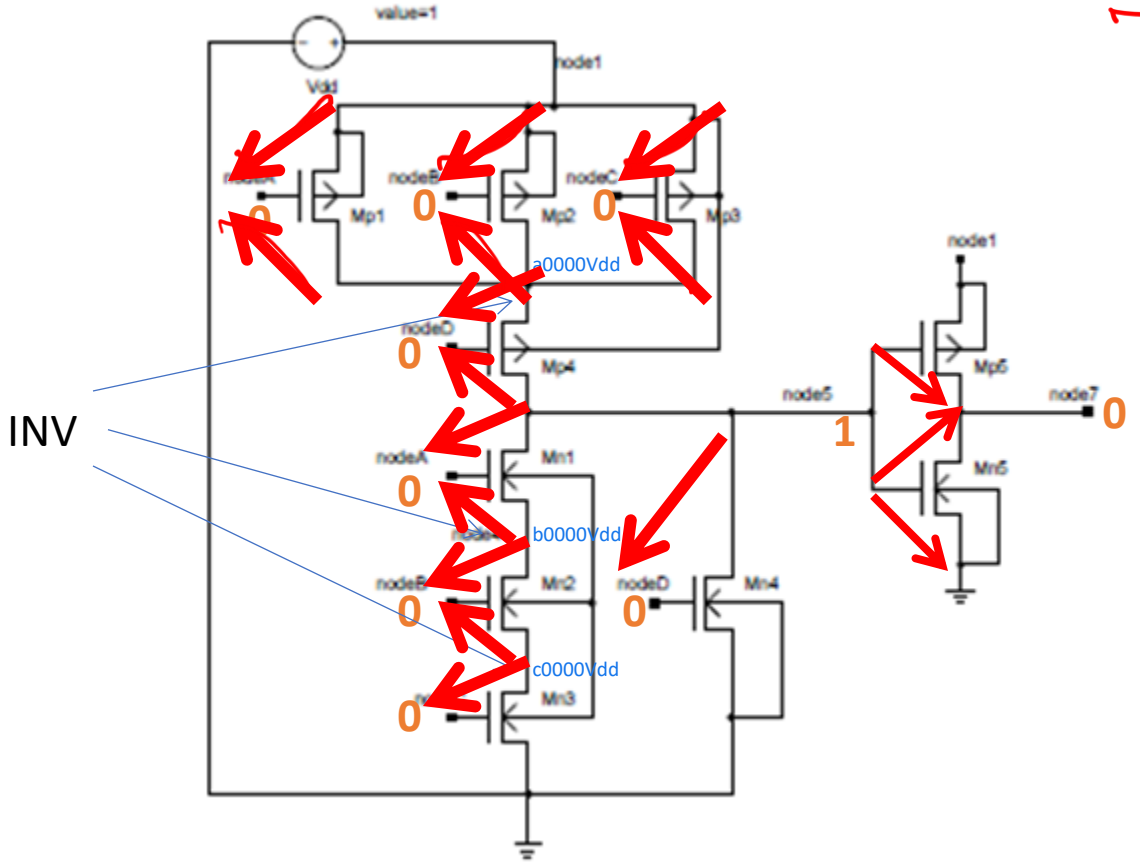
Sub-threshold leakage

Let’s take an example of input pattern is ABCD = 0000



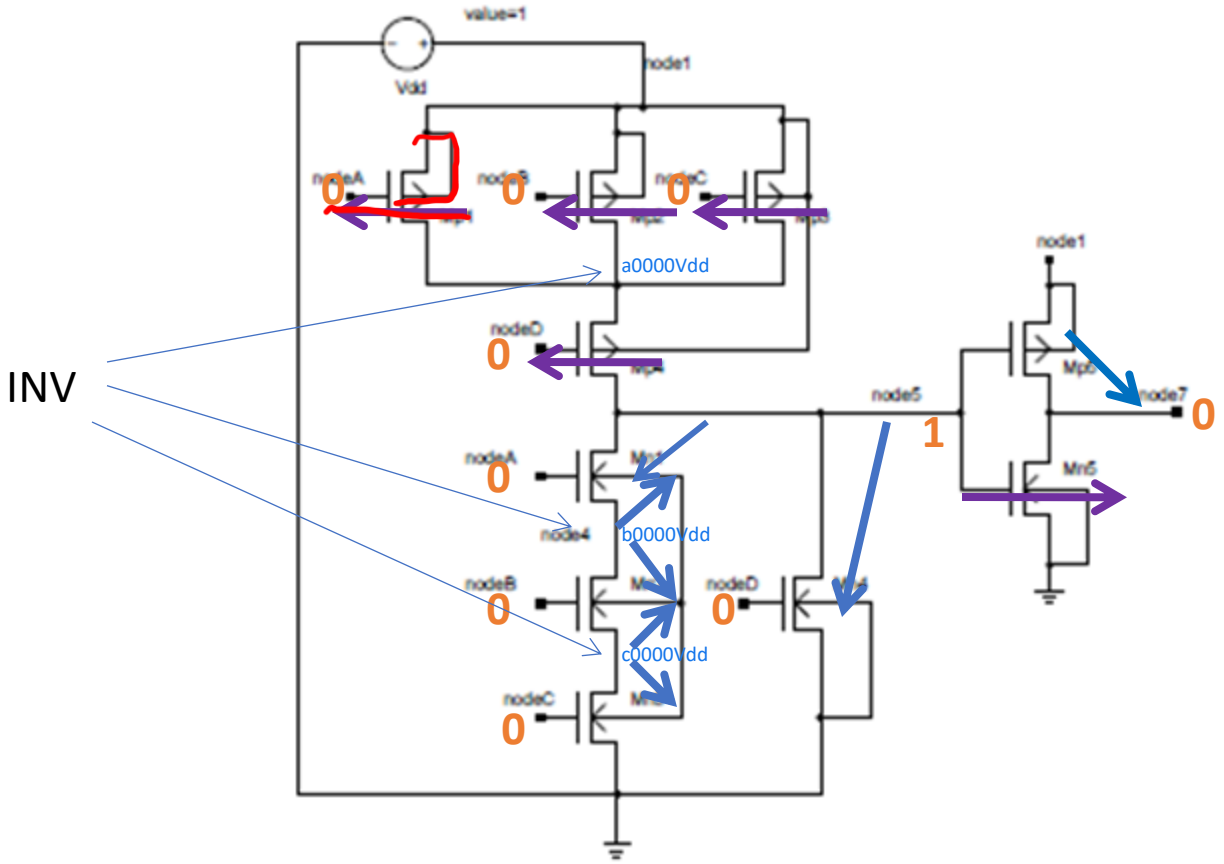
Gate Leakage

ABCD = 0000



Body Leakage

ABCD = 0000

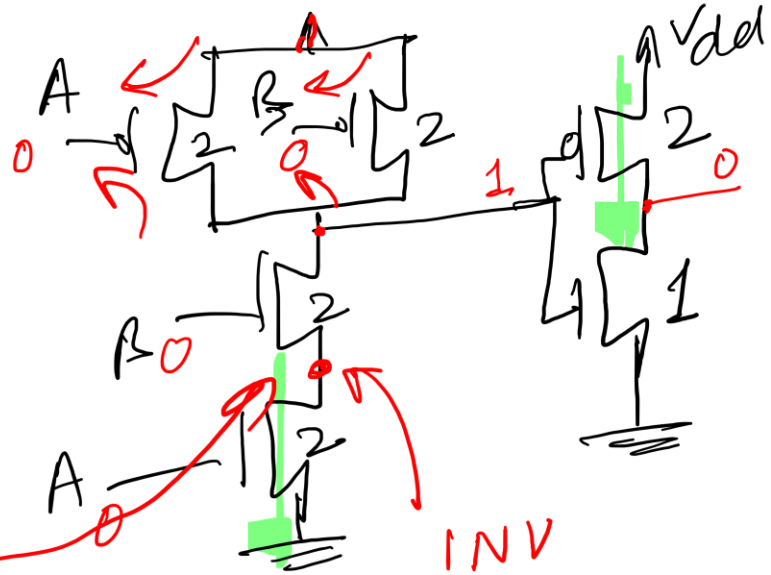


Leakage estimation equations: example_AND2

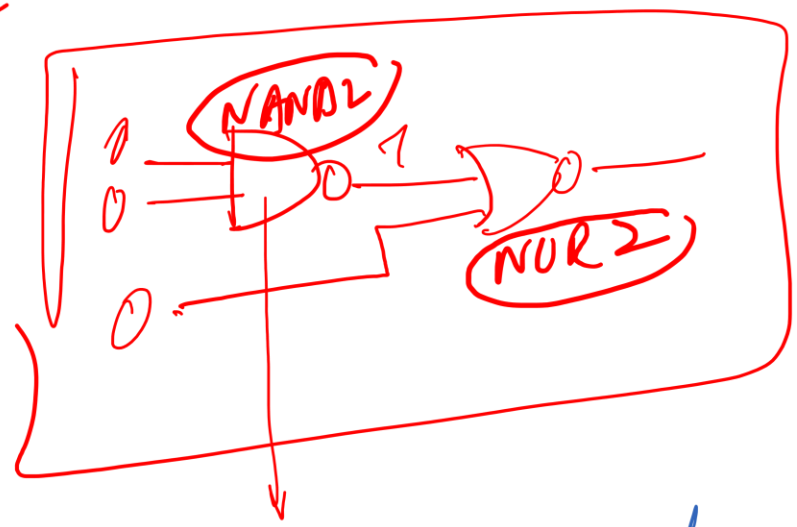
```
when "00" => PrintLeakages
(Isub(Wmin,2*NWmin,a00Vdd, temperature)+
Isub(Wmin,2*NWmin,-Vdd, temperature),

2.0*Ibody(Wmin,2*NWmin,+a00Vdd,0.0,temperature)+
Ibody(Wmin,2*NWmin,Vdd,0.0,temperature)+
2.0*Ibody(Wmin,2*NWmin,0.0,-Vdd,temperature)+
Ibody(Wmin,NWmin,0.0,Vdd,temperature)+Ibody(Wmin,2*NWmin,-Vdd,0.0,temperature),

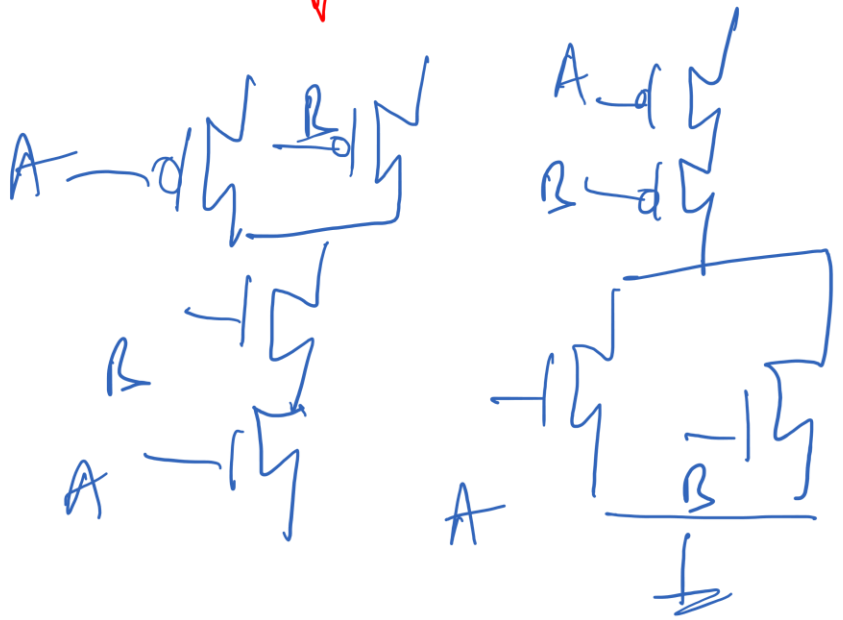
4.0*Igate(1,Wmin,2*NWmin,Vdd,temperature)+
2.0*Igate(0,Wmin,2*NWmin,a00Vdd,temperature)+
Igate(0,Wmin,2*NWmin,Vdd,temperature)+
2.0*Igate(0,Wmin,NWmin,-Vdd,temperature)+
Igate(1,Wmin,2*NWmin,-Vdd,temperature)
);
```



example



for all i/p Patterns



NOT → 0 → 1

NAND2 → 00 → 11

NAND3 → 000 → 111